Course Code: CSE475
Course Title: Machine Learning

Section: 06

[Semester: Summer 2026]

Project Task: 01
Project Title: Explainable ML model for Skin-Cancer Detection

Submitted by (Group-2):

Student Name	Student ID
Md. Rahat Khan	2021-1-60-090
Fahim Hossain	2022-2-60-142
Md Ahsiul Karim	2022-3-60-074
Jannatul Alam Shifa	2022-2-60-147

Submitted To:

Dr. Raihan Ul Islam
Associate Professor, Department of Computer Science and Engineering,

Related Work Table:

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
DSCC_Net: Multi-Classification Deep Learning Models for Diagnosing of Skin Cancer Using Dermoscopic Images	Deep Learning Models for Diagnosing of Skin Cancer Using Dermoscopic Images ISIC 2020: challenge2020.isicarchive.com HAM10000 URL: NR DermIS: dermis.net	4300 images; 4 classes. BCC 510, MEL 1686, MN 2007, SCC 97. After SMOTE Tomek: 8012 images. Train/validation/test split: NR.	Custom DSCC_Net CNN; SMOTE Tomek; Grad-CAM. Compared with ResNet-152, VGG-16/19, MobileNet, EfficientNet-B0 and Inception-V3	Accuracy 94.17% Precision 94.28% Recall 93.76% F1 93.93% AUC 99.43%	Can a custom CNN classify MEL, BCC, SCC and MN better than common deep networks while addressing class imbalance?	Pros: • High four-class accuracy. • Imbalance handling and XAI. Cons: • Fair-toned images only. • Split and key hyperparameters NR. • Some reported metrics conflict across sections.	M. Tahir et al., "DSCC_Net: MultiClassification Deep Learning Models for Diagnosing of Skin Cancer Using Dermoscopic Images," Cancers, vol. 15, Art. 2179, 2023, doi: 10.3390/cancers15072179.
Deep Learning-Based Skin Diseases Classification using Smartphones	PAD-UFES-20 URL: NR Monkeypox Skin Lesion Dataset URL: NR	7 classes: ACK, BCC, MEL, NEV, SCC, SEK and MPX. Text reports 2298 PAD-UFES-20 + 102 MSLD images. Table reports MPX 1428, creating a conflict. Split: 80%/20%.	ImageNet transfer learning with seven lightweight CNNs. Selected ResNet-18 converted to TensorFlow Lite and deployed on Android. Training data augmentation used.	ResNet-18: 74.27% Precision 76.78% Recall 71.40% F1 73.63% Jaccard 59.68% EfficientNetB3: 75.88%	Can a compact pretrained CNN perform seven-class skindisease classification, including MPX, in real time on smartphones?	Pros: • Working Android deployment. • Speed and FPS tested on four phones. Cons: • Lower accuracy than other reviewed models. • MPX count conflict and possible source bias. • No prospective clinical validation.	I. Oztel, G. Y. Oztel, and V. H. Sahin, "Deep Learning-Based Skin Diseases Classification using Smartphones," Adv. Intell. Syst., vol. 5, Art. 2300211, 2023, doi: 10.1002/aisy.202300211.

Skin cancer classification using vision transformers and explainable artificial intelligence	ISIC: isic-archive.com Specific Kaggle subset; thirdparty URL omitted.	3297 images; 2 classes. Training: 2637 (1440 benign, 1197 malignant). Testing: 660 (360 benign, 300 malignant). SMOTE applied only to training data. Validation split: NR.	ViT-base/large; Swin-small/base/tiny; VGG19, ResNet18/50, DenseNet201 and MobileNetV2. ImageNet transfer learning. Grad-CAM, Grad-CAM+ + and Score-CAM.	Best: ResNet50 88.8% Precision 86.9% Sensitivity 88.6% F1 87.8% Specificity 88.9% ViT-base/large: 88.6%	How do ViT, Swin and CNN models compare for benignversus-malignant classification, and can CAM-based XAI explain the best model?	Pros: • Direct CNN-transformer comparison. • Three XAI methods. Cons: • Binary task and small dataset. • XAI not quantitatively validated. • Some heatmaps focus on irrelevant regions. • No external validation.	G. H. Dagnaw, M. El Mouhtadi, and M. Mustapha, "Skin cancer classification using vision transformers and explainable artificial intelligence," J. Med. Artif. Intell., vol. 7, Art. 14, 2024, doi: 10.21037/jmai-24-6.
Enhancing Dermatological Diagnostics with EfficientNet: A Deep Learning Approach	Authors' clinical collection: private, URL NR ISIC 2019: gallery.isicarchive.com	8222 original images; 6 classes. MEL 1655, BCC 1811, BKL 1663, MN 1686, SCC 606, AK 801. After augmentation: 2144/class. Training 6578; validation + test 1644; separate split NR.	Customized ImageNet-pretrained EfficientNetB3 with batch normalization, 256-unit dense layer, dropout 0.45 and Softmax. Adamax, categorical crossentropy and augmentation.	4 classes: 95.4% validation; test 0.96 6 classes: 88.8% validation; test 0.89 AUC: 0.98-1.00 (4 classes); 0.93-1.00 (6 classes)	Can a customized EfficientNetB3 provide accurate and computationally efficient classification of four and six skinlesion categories?	Pros: • Strong four-class performance. • Class-wise metrics, ROC and confusion matrices. Cons: • Accuracy drops with AK and SCC. • SCC recall 0.54, F1 0.61. • Separate validation/test split NR. • No independent external validation.	I. Manole, A.-I. Butacu, R. N. Bejan, and G.-S. Tiplica, "Enhancing Dermatological Diagnostics with EfficientNet: A Deep Learning Approach," Bioengineering, vol. 11, Art. 810, 2024, doi: 10.3390/bioengineering 11080810.

EfficientSkinDis: An EfficientNet-based classification model for a large manually curated dataset of 31 skin diseases	Atlas Dermatology: atlasdermatologico.com.br ISIC: isic-archive.com Released dataset: Google Drive link in paper	4910 images; 31 classes; original class sizes 80-478. Final split: 3916 training and 994 testing. Augmented training count conflicts: 44,918 in Table 2 vs 37,551 in text.	Transfer-learned EfficientNet-B0- B6, ResNet-50/101/152 and VGG-16/19. Best: fine-tuned EfficientNet-B2. Six augmentations, 10-fold crossvalidation and Streamlit deployment.	EfficientNet-B2: 87.15% Precision 0.87 Recall 0.87 F1 0.87 Mean AUC 0.91 10-fold range: 0.8511-0.8715	Which transfer-learned CNN performs best for 31 diseases, and how do split choice and augmentation affect performance?	Pros: • Widest disease coverage. • Twelve-model comparison and 10-fold validation. • Public web demonstration. Cons: • Source and acquisition bias possible. • Patient-level split and label verification NR. • Model selection relied on test accuracy. • Augmented counts conflict.	A. Rafay and W. Hussain, “EfficientSkinDis: An EfficientNetbased classification model for a large manually curated dataset of 31 skin diseases,” Biomed. Signal Process. Control, vol. 85, Art. 104869, 2023, doi: 10.1016/j.bspc.2023.104869.

Title: Exploratory Data Analysis of the HAM10000 skin cancer dataset.

1. **Introduction:** Exploratory Data Analysis (EDA) is an important step before developing any machine learning or deep learning model. It helps understand the dataset, identify missing values, analyze data distributions, detect class imbalance, and gain insights that guide preprocessing and model development.

In this project, EDA was performed on the **HAM10000 (Human Against Machine with 10000 training images)** dataset, which contains dermoscopic images of skin lesions belonging to seven different disease categories.

2. **Dataset Description:** The HAM10000 dataset contains **10,015 dermoscopic images** of pigmented skin lesions along with metadata describing each image.

Dataset Information:

Feature	Description
Total Images	10,015
Number of Classes	7
Metadata File	HAM1000_metadata.csv
Image Format	JPG
Image Type	RGB

Metadata Attributes:

- Lesion ID
- Image ID
- Diagnosis (dx)
- Diagnosis Type (dx_type)
- Age
- Sex
- Localization

	lesion_id	image_id	dx	dx_type	age	sex	localization
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear

3. **Objectives of the EDA:** The main objectives of this analysis were:

- To understand the structure of the dataset.
- To identify missing values.
- To analyze the distribution of disease classes.
- To examine patient demographics such as age and gender.
- To study lesion locations on the human body.
- To inspect sample demoscopic images.
- To verify image dimensions before preprocessing.

4. **Data Loading and Inspection:** The dataset was loaded using the Pandas library. Basic information such as the number of rows, columns, data types, and descriptive statistics was examined.

The dataset contains both categorical and numerical attributes. Most columns were complete, although a small number of missing values were observed in the **Age** column.

5. **Missing Value Analysis:** Missing values were checked using the isnull() function. We observed:
- The **Age** column contains some missing values.
- All other metadata fields are largely complete.

Handling missing values is necessary before model training to avoid inconsistencies.

6. **Disease Class Distribution:** The frequency of each disease class was visualized using a count plot.

The seven disease classes are:

- Actinic Keratoses (akiec)
- Basal Cell Carcinoma (bcc)
- Benign Keratosis (bkl)
- Dermatofibroma (df)
- Melanoma (mel)
- Melanocytic Nevi (nv)
- Vascular Lesions (vasc)

Observation: The dataset is **highly imbalanced**. The **Melanocytic Nevi (nv)** class contains the largest number of images, while **Dermatofibroma (df)** and **Vascular Lesions (vasc)** contain comparatively few samples.

This class imbalance should be considered during model training because it may bias the classifier toward the majority class.

7. **Gender Distribution:** The gender distribution of patients was analyzed using a count plot.

Observation: Both male and female patients are represented in the dataset. The distribution is relatively balanced, although a small number of records have unknown gender information.

8. **Age Distribution:** The age distribution was visualized using a histogram. Most patients are between 40 and 70 years of age. Very young patients are comparatively rare. This suggests that skin lesions are more frequently recorded among middle-aged and older adults.

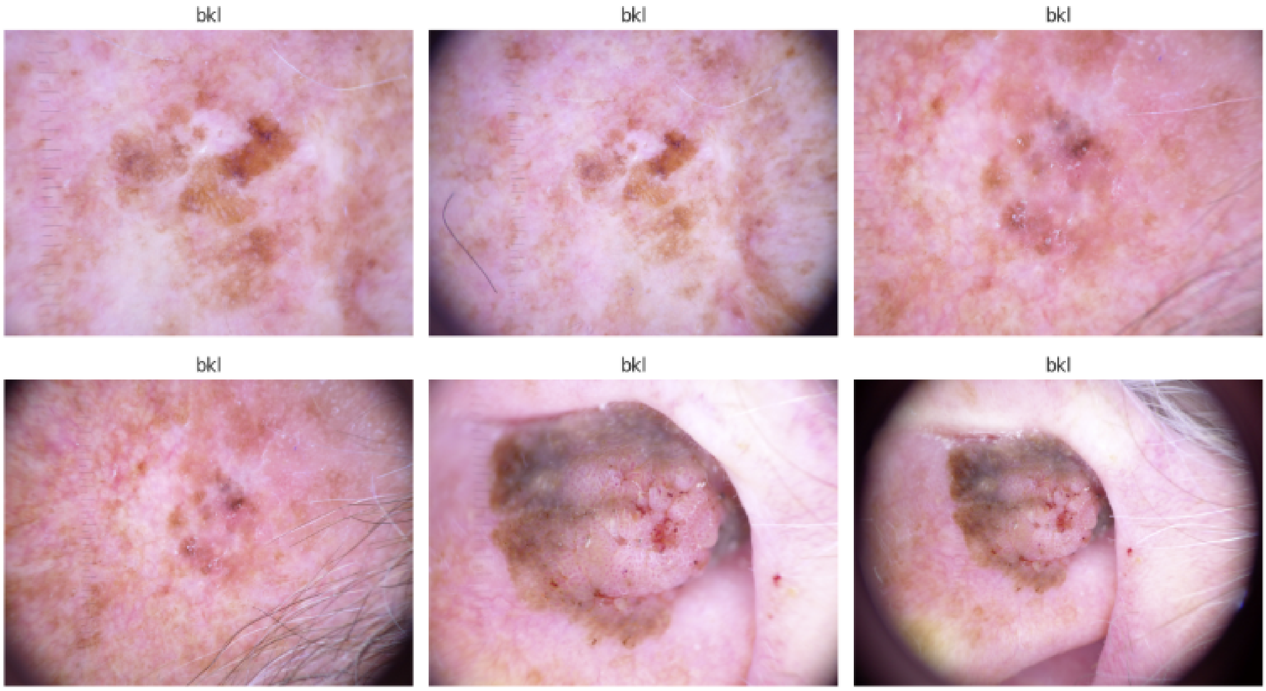
9. **Lesion Localization:** The body locations of skin lesions were analyzed using a horizontal bar chart. Common lesion locations include: Back, Lower Extremity, Trunk, Upper Extremity. The **back** contains the highest number of recorded lesions, followed by the lower extremities.

10. **Diagnosis Type:** Different diagnosis methods were analyzed. The metadata includes diagnosis confirmation methods such as:

- Histopathology
- Follow-up
- Consensus
- Confocal Microscopy

Most diagnoses were confirmed using **histopathology**, making the dataset reliable for supervised learning tasks.

11. **Sample Image Visualization:** Several dermoscopic images were displayed with their corresponding disease labels (ID = ISIC_0024306 to ISIC_0024311).



The above sample images show noticeable differences in color, texture, size, and lesion shape. However, visual similarities between some disease categories indicate that automatic classification is a challenging task.

12. **Image Size Analysis:** The dimensions of all images were extracted and summarized. We observed that the images have consistent dimensions throughout the dataset. Uniform image sizes simplify preprocessing because additional resizing requirements are minimal.

13. **Key Findings:** The exploratory analysis revealed several important characteristics of the dataset:

- The dataset contains **10,015 dermoscopic images** across **seven disease classes**.
- A small number of missing values exist in the **Age** column.
- The dataset is significantly **class imbalanced**, with the **Melanocytic Nevi (nv)** class dominating.
- Most patients are middle-aged or older.
- Skin lesions most frequently occur on the back and lower extremities.
- Images have consistent dimensions, making preprocessing straightforward.
- Histopathology is the most common diagnosis method, increasing the credibility of the dataset

14. **Conclusion:** The Exploratory Data Analysis provided a comprehensive understanding of the HAM10000 dataset and identified several important characteristics relevant to skin cancer classification. The analysis highlighted class imbalance, missing values, patient demographics, lesion locations, and image properties. These findings will guide the subsequent preprocessing, data augmentation, and model development stages. Overall, the dataset is well-suited for developing deep learning models for automated skin lesion classification, provided that appropriate techniques are applied to address class imbalance and optimize model performance.